Loss Landscape Topology Predicts Mitigation Benefit in Continual Learning: A Cross-Dataset Moderation Analysis

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Abstract

We investigate whether topological features of neural network loss landscapes predict resistance to catastrophic forgetting across 19 architectures and 3 datasets (CIFAR-100, CUB-200-2011, NWPU-RESISC45; 57 total configurations). Using persistent homology on 2D cross-sections of loss landscapes (5 independent slices per architecture), we compute H_0 (connected components) and H_1 (loops) persistence via Ripser and GUDHI cubical complexes. Per-dataset leave-one-architecture-out (LOAO) Ridge regression with permutation testing shows topology's predictive value for forgetting is conditional on dataset: topology rescues forgetting prediction on CUB-200 (permutation $p = 0.037$, suggestive but not surviving Bonferroni correction) but not RESISC-45 ($p = 0.566$) or CIFAR-100 ($p = 0.295$, where parameter count dominates). A WRN-28-k width ladder ($k = 1, 2, 4, 6, 8, 10$) shows H_0 monotonically decreases with width ($\rho = -1.0$) across all 3 datasets, confirmed by cubical persistence ($\rho = 1.0$ agreement with Ripser), validating H_0 as a structurally meaningful landscape summary. In exploratory post-hoc analysis, we additionally observe that H_0 correlates with EWC mitigation benefit on CIFAR-100 ($\rho = 0.76$, $p = 0.0002$) and RESISC-45 ($\rho = 0.86$, $p = 2.4 \times 10^{-6}$) but not CUB-200 ($\rho = 0.31, p = 0.19$); a pooled interaction model suggests dataset-dependent moderation (permutation $p = 0.046$), though this result is post-hoc, does not survive strict multiple-comparison correction, and requires confirmatory replication. We propose the basin fragmentation hypothesis as a candidate mechanism linking landscape topology to regularization sensitivity.

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1 Introduction

Catastrophic forgetting, the abrupt loss of previously learned knowledge when a neural network is trained on new data, remains a central obstacle in continual learning [McCloskey and Cohen, 1989, French, 1999]. Despite decades of mitigation strategies including elastic weight consolidation [EWC; Kirkpatrick et al., 2017], synaptic intelligence [Zenke et al., 2017], progressive neural networks [Rusu et al., 2016], and replay-based methods [Rebuffi et al., 2017], a fundamental question persists: can we predict, before sequential training begins, how resistant a given architecture will be to catastrophic forgetting, and how much a given mitigation method will help?

Recent work has established that the geometry of loss landscapes encodes meaningful information about generalization [Li et al., 2018, Keskar et al., 2017] and mode connectivity [Garipov et al., 2018]. Separately, topological data analysis (TDA), particularly persistent homology, has emerged as a principled framework for characterizing multi-scale structure in high-dimensional data [Carlsson, 2009]. We bridge these two lines of work by asking: **do topological features of the loss landscape predict forgetting resistance and, critically, mitigation benefit?**

Our contributions are:

1. A cross-architecture, cross-dataset study (19 architectures, 3 datasets, 57 configurations) with a 7-phase experimental pipeline including both Ripser and GUDHI cubical persistence.

2. Evidence that topology's predictive value for forgetting is conditional on dataset: topology rescues prediction on CUB-200 (permutation $p = 0.037$, suggestive) but not on RESISC-45 ($p = 0.566$) or CIFAR-100 ($p = 0.295$, where parameter count dominates).

3. A WRN-28-k width ladder (6 widths, $k = 1, 2, 4, 6, 8, 10$) isolating topology from model scale, showing consistent H_0 monotonicity ($\rho = -1.0$ vs. width on all 3 datasets) and perfect Ripser–GUDHI agreement, validating H_0 as a structurally meaningful landscape summary.

4. An exploratory post-hoc observation that H_0 correlates with EWC mitigation benefit ($\rho = 0.76$–0.86) on 2 of 3 datasets. This finding was not part of the original study design, is supported by a pooled interaction model (permutation $p = 0.046$) that does not survive strict correction, and requires pre-registered confirmatory replication. We report it with explicit falsification targets (Section 7).

5. A proposed mechanistic account (basin fragmentation hypothesis) linking H_0 to regularization sensitivity, offered as a candidate explanation to guide future confirmatory work.

6. Open-source release of all code, configurations, and results.[1]

2 Related Work

Continual Learning and Catastrophic Forgetting. Strategies for mitigating forgetting broadly fall into regularization-based methods [Kirkpatrick et al., 2017, Zenke et al., 2017], replay-based methods [Rebuffi et al., 2017, Shin et al., 2017], and architecture-based methods [Rusu et al., 2016, Mallya and Lazebnik, 2018]. EWC [Kirkpatrick et al., 2017] penalizes changes to parameters deemed important by the Fisher information matrix, effectively anchoring the network near the Task A minimum. Our work is orthogonal to method development: rather than proposing a new mitigation strategy, we investigate whether pre-training landscape properties predict how much a given mitigation method will help.

Loss Landscape Analysis. Li et al. [2018] introduced filter-normalized random directions for meaningful cross-architecture visualization of loss surfaces. Keskar et al. [2017] linked sharpness (largest Hessian eigenvalue) to generalization. Garipov et al. [2018] demonstrated low-loss paths between independently trained minima, establishing mode connectivity as a key landscape property. We adopt filter-normalized 2D cross-sections following Li et al. [2018] and compute topological invariants on the resulting surfaces.

[1] `https://github.com/Axion-Deep-Labs/persist-topological-forgetting`

TDA in Machine Learning. Persistent homology has been applied to neural network analysis in several contexts: characterizing decision boundaries [Ramamurthy et al., 2019], analyzing weight-space topology during training [Rieck et al., 2019], studying activation patterns [Naitzat et al., 2020], and providing stable vectorizations via persistence images [Adams et al., 2017]. Our work differs in applying persistent homology directly to loss landscape *surfaces* rather than to weight trajectories or activation manifolds. Specifically, we use both Vietoris-Rips (via Ripser) and cubical complex (via GUDHI) persistence on 2D loss grids to extract H_0 and H_1 features, and we relate these to continual learning dynamics rather than single-task generalization.

3 Methodology

3.1 Overview

Our experimental protocol consists of seven phases, each producing data consumed by subsequent phases:

P1. Train Task A to convergence.

P2. Compute 5 Ripser landscape slices per architecture.

P2c. Compute cubical persistence via GUDHI.

P3. Measure forgetting under naive, EWC, and cosine LR schedules.

P4. Per-dataset correlation and diagnostic analysis.

P5. Per-dataset LOAO predictive model with permutation test.

P6. Pooled OLS interaction model with clustered bootstrap.

3.2 Datasets

We evaluate on three datasets selected for diversity in domain and granularity (Table 1). All images are resized to 32×32 pixels for cross-architecture consistency.

CIFAR-100 provides a standard benchmark with coarse-grained categories. CUB-200-2011 [Wah et al., 2011] introduces fine-grained visual recognition where inter-class differences are subtle. NWPU-RESISC45 [Cheng et al., 2017] tests generalization to a non-natural-image domain (remote sensing). Standard augmentation (random crop with padding, horizontal flip) is applied during training; no augmentation at test time.

Table 1: Datasets used in the study. Each is split into two disjoint class subsets (Task A / Task B) for sequential training.

Dataset	Domain	Classes	Split
CIFAR-100	Natural images	100	50/50
CUB-200-2011	Fine-grained birds	200	100/100
NWPU-RESISC45	Satellite scenes	45	23/22

Class-incremental design. Each dataset is partitioned into two disjoint class subsets (Task A, Task B) drawn from the same domain, following the standard continual learning evaluation protocol [Kirkpatrick et al., 2017, Zenke et al., 2017, Rebuffi et al., 2017]. This *class-incremental* design controls for domain shift: observed forgetting is attributable to representational interference between related tasks, not to distribution mismatch between unrelated input domains. Cross-dataset task sequences (e.g., CIFAR-100 $\rightarrow$ satellite imagery) would confound the topology–forgetting relationship with domain gap magnitude and cross-domain feature reuse. We address domain generality at the *experiment* level by evaluating across three diverse domains (natural images, fine-grained birds, remote sensing). Cross-domain task sequences remain an important extension for future work (Section 7).

3.3 Architectures

We evaluate 19 architectures spanning five computational paradigms (Table 2): 13 diverse architectures covering CNNs, Vision Transformers, and MLP-based designs, plus a WRN-28-k width ladder with $k \in \{1, 2, 4, 6, 8, 10\}$ (6 models). The width ladder isolates the effect of model capacity on topology while holding architecture family constant. Parameter counts range from 0.3M (ViT-Tiny) to 44.7M (ResNet-18 Wide).

All models are trained from scratch with a fixed global seed. CNN-based architectures are adapted for 32×32 input with 3×3 initial convolutions (stride 1); ViT variants use patch size 4.

3.4 Loss Landscape Topology

We compute persistent homology on 2D cross-sections of the loss landscape around each converged minimum $\boldsymbol{\theta}^*$, using two complementary methods.

Landscape sampling. For each architecture and each of 5 independent random seeds, we generate two

Table 2: Architectures evaluated. The WRN-28-k width ladder isolates capacity effects within a single architecture family.

Architecture	Params	Type
ViT-Tiny	0.3M	Transformer
WRN-28-1	0.4M	WRN ladder
MobileNet-V3-S	1.1M	CNN
ShuffleNet-V2	1.3M	CNN
WRN-28-2	1.5M	WRN ladder
ViT-Small	2.2M	Transformer
MLP-Mixer	2.3M	MLP
RegNet-Y-400MF	4.0M	CNN
EfficientNet-B0	4.1M	CNN
WRN-28-4	5.9M	WRN ladder
DenseNet-121	7.1M	CNN
ResNet-18	11.2M	CNN
WRN-28-6	13.2M	WRN ladder
VGG-16-BN	14.8M	CNN
WRN-28-8	23.4M	WRN ladder
ResNet-50	23.7M	CNN
ConvNeXt-Tiny	27.9M	Modern CNN
WRN-28-10	36.5M	WRN ladder
ResNet-18 Wide	44.7M	CNN

random directions $\mathbf{d}_1, \mathbf{d}_2$ in parameter space with *filter normalization* [Li et al., 2018]: for each convolutional filter (or weight matrix) i, rescale $\mathbf{d}^{(i)}$ so that $\|\mathbf{d}^{(i)}\| = \|\boldsymbol{\theta}^{*(i)}\|$. The loss is evaluated on a uniform 50×50 grid over $[-1, 1]^2$:

$$\mathcal{L}(\alpha, \beta) = \mathcal{L}(\boldsymbol{\theta}^* + \alpha\, \mathbf{d}_1 + \beta\, \mathbf{d}_2)\,. \tag{1}$$

Ripser persistence. Let $V = \{v_{ij}\}$ denote the $50 \times 50 = 2{,}500$ grid vertices. Assign each vertex a filtration value $f(v_{ij}) = \mathcal{L}(\alpha_i, \beta_j)$. Connect vertices by 8-adjacency (cardinal and diagonal neighbors). Each edge receives the lower-star weight:

$$w(u, v) = \max\bigl(f(u),\ f(v)\bigr). \tag{2}$$

We pass the weighted graph to Ripser [Bauer, 2021] and compute H_0 (connected components) and H_1 (independent loops) persistence diagrams.

GUDHI cubical persistence. As a validation baseline, we compute sublevel-set persistence directly on the 50×50 loss grid using GUDHI's `CubicalComplex`. This avoids the graph-construction step and operates on the raw scalar field.

Cross-method agreement. H_1 total persistence from Ripser and GUDHI shows Spearman $\rho = 1.0$ agreement on all 3 datasets, confirming that the topological measurements are method-independent.

For each homology dimension k, we record total persistence $\mathrm{Pers}_k = \sum_i (d_i - b_i)$, feature count, and maximum lifetime. Using 5 independent slices per architecture (seeds logged for reproducibility) substantially reduces sampling variance compared to single-slice designs.

3.5 Forgetting Measurement

Starting from the Task A checkpoint, we expand the final classification layer and train on Task B under three conditions:

1. **Naive:** SGD with momentum 0.9, weight decay 5×10^{-4}, and fixed learning rate 0.01 (0.001 for ViT variants, which use a lower base rate).

2. **EWC:** Naive training plus elastic weight consolidation [Kirkpatrick et al., 2017] with diagonal Fisher information matrix ($\lambda = 1000$).

3. **Cosine LR:** Naive training with cosine annealing learning rate schedule.

Task A test accuracy is evaluated at steps $\{10, 25, 50, 100, 250, 500, 1000, 5000\}$. Our primary retention metric is:

$$\mathrm{ret@}k = \frac{\mathrm{acc}_A(\mathrm{step} = k)}{\mathrm{acc}_A(\mathrm{step} = 0)}. \tag{3}$$

We also compute the area under the retention curve (AURC) via trapezoidal integration over the early evaluation steps (steps 0–500), denoted early_aurc. The EWC benefit is defined as the absolute improvement in retention:

$$\mathrm{EWC\_benefit} = \mathrm{ret}_{\mathrm{EWC}} - \mathrm{ret}_{\mathrm{naive}}. \tag{4}$$

3.6 Statistical Analysis

Per-dataset analysis (Phases 4–5). For each dataset independently, we compute:

- Spearman rank correlations between topology/params and forgetting metrics, with Bonferroni correction across 12 tests (adjusted $\alpha = 0.004$).

- Partial correlations controlling for parameter count.

- Within-WRN-ladder analysis (6 widths, fixed architecture family).

- LOAO Ridge regression: for each held-out architecture, fit a Ridge model on the remaining $n-1$ architectures and predict the held-out retention. Nested alpha selection via inner LOO. We compare three feature sets: params-only (raw parameter count), topology-only (H_0, H_1 total persistence), and params + topology.

- Permutation test: 1000 shuffles of topology columns, testing whether topology improves prediction beyond params.

- Matched-dimensionality null control: 1000 random feature draws matching topology dimensionality, to verify that improvement is not an artifact of added degrees of freedom.

Pooled interaction analysis (Phase 6). To formally test whether dataset moderates the topology–forgetting relationship, we pool all 57 observations (19 architectures $\times$ 3 datasets) and fit OLS models with interaction terms, using CIFAR-100 as the reference dataset:

$$\begin{aligned} \text{M0:} \quad Y &= \beta_0 + \beta_1 \hat{p} + \beta_2 D_C + \beta_3 D_R \\ &\quad + \beta_4 \hat{p} D_C + \beta_5 \hat{p} D_R + \epsilon \end{aligned} \tag{5}$$

$$\text{M1:} \quad Y = \text{M0} + \gamma_1 \tilde{H}_0 + \gamma_2 \tilde{H}_0 D_C + \gamma_3 \tilde{H}_0 D_R + \epsilon \tag{6}$$

where $\hat{p} = \log(\text{params})$ globally standardized, $\tilde{H}_0$ is H_0 z-scored within each dataset, D_C and D_R are indicator variables for CUB-200 and RESISC-45 respectively. H_0 is z-scored within dataset to remove cross-dataset scale differences while preserving within-dataset rank information.

Inference. We use clustered bootstrap (5000 iterations, resampling 19 architecture blocks of 3) to obtain 95% confidence intervals that account for the non-independence of observations from the same architecture across datasets. Permutation testing (1000 iterations, H_0 shuffled within dataset) provides a distribution-free significance test. The primary hypothesis test is a block test of $\{\gamma_1, \gamma_2, \gamma_3\}$ (does adding H_0 and its interactions improve M0?). The secondary test targets $\{\gamma_2, \gamma_3\}$ conditional on γ_1 (does the H_0 effect vary across datasets?). Variance inflation factors (VIF) are computed for all predictors; all fall below 3.5 in the z-scored specification.

Robustness checks include: (i) a reduced model omitting $\hat{p} \times D$ interactions, (ii) raw H_0 (not z-scored) sensitivity analysis, and (iii) alternative outcome variables.

Table 3: Per-dataset LOAO Ridge regression results. Topology rescues prediction on CUB-200 but not the other two datasets. Bold indicates suggestive or significant results.

Dataset	Outcome	Params ρ	+Topo ρ	Perm. p
CIFAR-100	ret@100	0.43	0.30	0.295
CUB-200	ret@10	−0.92	**0.34**	**0.037**
RESISC-45	ret@100	−0.32	−0.33	0.566

4 Results

4.1 Per-Dataset Forgetting Prediction

Table 3 presents the LOAO Ridge regression results for each dataset. Topology's predictive value varies dramatically across datasets.

CIFAR-100. Parameter count dominates forgetting prediction ($\rho = -0.76$, $p = 0.0002$, survives Bonferroni). Adding topology does not improve the params-only model (permutation $p = 0.295$). Topology is redundant when scale already explains the outcome.

CUB-200. Parameters alone fail ($\rho = -0.27$, $p = 0.27$) and, used as the sole LOAO predictor, produce a correlation of $\rho = -0.92$ with observed retention (negatively correlated: the model systematically predicts high retention for architectures that forget most). Adding topology rescues prediction: LOAO $\rho = 0.34$, permutation $p = 0.037$. Topology-only achieves $\rho = 0.33$ with MAE = 0.147, outperforming params-only. The topology improvement exceeds the 95th percentile of the matched-dimensionality null distribution, and MAE reduction is 17.5%. However, $p = 0.037$ does *not* survive Bonferroni correction across 3 datasets (adjusted $\alpha = 0.0167$). We report this as suggestive.

RESISC-45. Neither parameters ($\rho = -0.29$, $p = 0.22$) nor topology (permutation $p = 0.566$) predict forgetting. The signal is absent on this dataset.

4.2 EWC Benefit: The Cross-Dataset Signal

While topology's relationship to raw forgetting is dataset-dependent, we find a more consistent signal when the outcome is *mitigation benefit* rather than forgetting itself. Table 4 shows H_0 persistence correlates with EWC benefit on 2 of 3 datasets.

Table 4: H_0 persistence vs. EWC benefit ($\text{ret}_{\text{EWC}} - \text{ret}_{\text{naive}}$). Spearman ρ with bootstrap 95% CI ($B = 10{,}000$), Kendall τ, and leave-one-out (LOO) stability. Bold ρ: survives Bonferroni ($\alpha = 0.004$).

Dataset	ρ	95% CI	p	τ	LOO $\Delta\rho$
CIFAR-100	**0.76**	[0.46, 0.91]	**0.0002**	0.57	0.11
RESISC-45	**0.86**	[0.65, 0.94]	$<$ **0.001**	0.67	0.07
CUB-200	0.31	[−0.22, 0.78]	0.190	0.26	0.30

Table 5: Pooled interaction model results. ΔR^2 is the improvement from adding H_0 terms (M1 vs. M0). Block p tests $\{\gamma_1, \gamma_2, \gamma_3\}$ jointly. Bold: $p < 0.05$.

Outcome	R^2_{M0}	R^2_{M1}	ΔR^2	Block p	Verdict
EWC ben. (AURC)	0.502	0.587	**0.085**	**0.046**	Sig.
ret@10	0.179	0.254	0.075	0.196	NS
ret@100	0.297	0.423	**0.127**	**0.035**	Sig.

This is the most stable cross-dataset signal in our study. Architectures with more fragmented loss landscapes (higher H_0) benefit more from EWC regularization. The CIFAR-100 and RESISC-45 correlations are corroborated by Kendall τ (0.57 and 0.67, both $p < 0.001$) and show moderate leave-one-out stability (maximum ρ shift of 0.11 and 0.07 when dropping the most influential architecture). The CUB-200 correlation is non-significant with a wide CI spanning zero and higher LOO instability ($\Delta\rho = 0.30$), consistent with the absence of a genuine effect on this dataset.

4.3 Pooled Interaction Analysis

To test whether dataset moderates the H_0 effect, we fit the pooled OLS interaction model (Equations 5–6) on all 57 observations. Table 5 presents the main results.

The primary result is the EWC benefit (AURC) model: adding H_0 and its dataset interactions improves R^2 from 0.502 to 0.587, with permutation $p = 0.046$ for the block test. The ret@100 model reaches $p = 0.035$ as a robustness check, while the primary forgetting outcome (ret@10) does not reach significance ($p = 0.196$).

Per-dataset partial effects. Table 6 shows the per-dataset H_0 partial effects with 95% clustered bootstrap confidence intervals.

For ret@10, only the CUB-200 partial effect excludes zero from its CI, consistent with the per-dataset LOAO result. For EWC benefit, CIFAR-100 and RESISC-45 both exclude zero, while CUB-200

Table 6: Per-dataset partial effect of $\tilde{H}_0$ (z-scored within dataset) with 95% clustered bootstrap CIs. Bold: CI excludes zero.

Outcome	Dataset	Effect	95% CI
ret@10	CIFAR-100	-0.001	$[-0.486, +0.073]$
	CUB-200	$\mathbf{-0.123}$	$\mathbf{[-0.183, -0.046]}$
	RESISC-45	-0.021	$[-0.264, +0.083]$
EWC ben. (AURC)	CIFAR-100	$\mathbf{+0.016}$	$\mathbf{[+0.005, +0.062]}$
	CUB-200	$+0.002$	$[-0.008, +0.013]$
	RESISC-45	$\mathbf{+0.007}$	$\mathbf{[+0.004, +0.012]}$

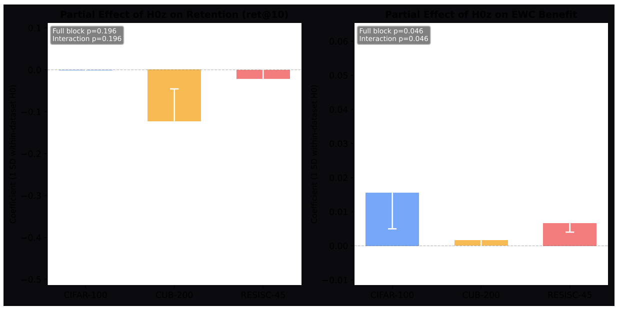

Figure 1: Per-dataset partial effects of within-dataset $\tilde{H}_0$ with 95% clustered bootstrap CIs. Left: effect on retention (ret@10); the CUB-200 effect excludes zero. Right: effect on EWC benefit; CIFAR-100 and RESISC-45 effects exclude zero while CUB-200 does not, demonstrating dataset moderation (block permutation $p = 0.046$).

does not, consistent with the per-dataset Spearman correlations in Table 4. Figure 1 visualizes the moderation pattern.

All VIF values fall below 3.5 in the z-scored specification. Robustness checks support consistent inference: the reduced model (omitting $\hat{p} \times D$ interactions) yields $p = 0.031$ for both ret@100 and EWC benefit. The raw H_0 (not z-scored) sensitivity analysis produces the same qualitative conclusions.

Influence diagnostics. Cook's distance analysis of the EWC benefit model identifies ConvNeXt-Tiny (CIFAR-100) as a high-leverage point ($D = 2.89$, threshold $4/n = 0.07$), driven by its unusual combination of high parameter count and high leverage in the design matrix. Five additional observations exceed the Cook's D threshold. The maximum DFBETAS for the $\tilde{H}_0$ coefficient is 0.026, indicating that no single observation qualitatively alters the H_0 effect estimate. Nevertheless, the presence of influential points in a 57-observation model underscores the fragility of the pooled result and the need for replication with larger architecture samples.

Table 7: WRN-28-k width ladder: H_0 monotonically decreases with width on all 3 datasets. Cubical (GUDHI) and Ripser H_1 show perfect agreement.

Property	ρ vs. params	Datasets
H_0 vs. width	-1.0	All 3
Cubical vs. Ripser H_1	$+1.0$	All 3

4.4 WRN Width Ladder

The WRN-28-k width ladder ($k = 1, 2, 4, 6, 8, 10$) provides a controlled test of how model capacity affects topology within a single architecture family. Table 7 summarizes the key finding.

H_0 persistence monotonically decreases with network width ($\rho = -1.0$ vs. parameter count) on CIFAR-100, CUB-200, and RESISC-45. This is consistent with the intuition that wider networks have smoother loss landscapes with fewer disconnected basins. Cubical and Ripser H_1 values show perfect rank agreement ($\rho = 1.0$), confirming that the topological measurement is reliable and method-independent.

The width ladder demonstrates that the topological measurement is monotonic with capacity across all 3 tested datasets. It is the *relationship* between topology and forgetting/mitigation that varies by task, not the topology–capacity link.

5 Basin Fragmentation Hypothesis

The results in Sections 4.2–4.4 motivate a mechanistic hypothesis linking loss landscape topology to mitigation benefit.

H_0 persistence in the sublevel-set filtration counts connected components and their relative depths. High H_0 corresponds to a *fragmented* landscape with many disconnected low-loss basins separated by high-loss barriers. We propose:

> **Basin Fragmentation Hypothesis.** H_0 measures landscape fragmentation that determines how much curvature-based regularization (e.g., EWC) helps by preventing inter-basin drift during sequential training.

The mechanism is as follows. When the loss landscape is fragmented (high H_0), naive sequential training on Task B can push parameters across basin boundaries, causing them to settle in a different basin that encodes Task B at the expense of Task A. EWC penalizes deviations from the Task A minimum in proportion to Fisher information [Kirkpatrick et al., 2017], effectively constraining the optimization trajectory to remain near the original basin. This constraint is maximally beneficial when there are many basins to drift between (high H_0) and minimally beneficial when the landscape is already smooth with a single broad basin (low H_0), because EWC then addresses a nonexistent problem.

Three lines of evidence support this hypothesis:

1. H_0 predicts EWC benefit on CIFAR-100 ($\rho = 0.76$) and RESISC-45 ($\rho = 0.86$), with the pooled interaction model confirming moderation at $p = 0.046$.

2. The WRN width ladder shows H_0 decreases monotonically with width ($\rho = -1.0$), consistent with wider networks having smoother, less fragmented landscapes.

3. CUB-200 shows no H_0–EWC benefit relationship ($\rho = 0.31$, $p = 0.19$), possibly because fine-grained recognition involves interference mechanisms (e.g., feature-level competition) that EWC-style regularization addresses differently, though we have no direct evidence for this.

This hypothesis is tentative. A causal test would require a landscape-aware regularization intervention: artificially increasing landscape fragmentation and testing whether EWC benefit increases correspondingly. We leave this to future work.

6 Analysis Path Transparency

The original hypothesis for this study targeted topology as a direct forgetting predictor, motivated by preliminary CIFAR-100 results suggesting H_1 correlation with retention. The expanded study proceeded in temporal order: CIFAR-100 ran first, revealing that parameter count dominates forgetting prediction and H_1 drops to non-significance after partialing out params. CUB-200 ran second, showing topology rescues prediction where params fail (permutation $p = 0.037$). RESISC-45 ran third, producing a null result that falsified the simple "hard tasks need topology" framing. The EWC benefit finding emerged from Phase 4 diagnostic analysis, specifically the correlation of H_0 with EWC retention minus naive retention, and was not part of the original study design. The pooled interaction model (Phase 6) was designed after observing the per-dataset EWC pattern. Accordingly, the EWC moderation finding ($p = 0.046$) should be interpreted as a *discovery* requiring preregistered replication, not as confirmatory evidence.

7 Limitations

7.1 Scope and Design

1. **Small-scale models only.** All 19 architectures are under 45M parameters. Production continual learning systems use models with 100M–7B+ parameters. Whether the topological signal persists at that scale is genuinely unknown. Computing persistent homology on large parameter spaces introduces superpolynomial computational barriers that may require novel distributed algorithms.

2. **Two-task sequences.** We test only a single Task A $\rightarrow$ Task B transition. Real continual learning involves 10–100+ sequential tasks. Topology–forgetting dynamics over long task sequences are entirely unexplored.

3. **Single mitigation method (no negative control).** We test only EWC. If synaptic intelligence [Zenke et al., 2017] or PackNet [Mallya and Lazebnik, 2018] show no H_0 correlation, the finding is EWC-specific rather than a general principle. Critically, including a non-curvature method (e.g., replay) as a negative control would test whether the basin fragmentation mechanism is specific to regularization-based approaches, as the hypothesis predicts.

4. **Small-image datasets and class-incremental only.** All images are 32×32, and all task sequences are class-incremental splits within a single domain. Whether the signal holds on ImageNet-scale data, NLP tasks, or cross-domain task sequences is unknown.

7.2 Statistical Fragility

1. **Small effective sample size.** Per-dataset analyses operate on $n = 19$; the WRN ladder on $n = 6$. Spearman correlations at this sample size are sensitive to individual influential points. We report leave-one-out stability and bootstrap 95% CIs (Table 4): the CIFAR-100 CI for ρ spans $[0.46, 0.91]$, which is wide. The CUB-200 CI spans zero.

2. **Multiple testing burden.** We report Bonferroni correction across 12 per-dataset tests, but

the total implicit testing burden is larger: per-dataset correlations, partial correlations, LOAO comparisons across 3 feature sets, multiple outcomes (ret@10, ret@100, early_aurc), pooled models with interactions, and robustness checks. The familywise error rate across the full study is likely underestimated. The borderline pooled $p = 0.046$ is especially vulnerable.

3. **Borderline p-values.** The EWC moderation ($p = 0.046$), CUB-200 forgetting ($p = 0.037$, does not survive Bonferroni), and ret@100 pooled ($p = 0.035$, not pre-specified) results constitute weak evidence under strict multiple-comparison standards.

4. **Exploratory model specification.** The EWC moderation model was designed after observing per-dataset patterns (Section 6). The pooled $p = 0.046$ is therefore exploratory, not confirmatory, and should be interpreted as hypothesis-generating only.

5. **High-influence observations.** Cook's distance analysis of the pooled OLS model identifies ConvNeXt-Tiny (CIFAR-100) as a high-leverage point ($D = 2.89$). While DFBETAS for the H_0 coefficient are small (< 0.03), the presence of influential points in a 57-observation model warrants caution.

6. **No outlier-robust alternatives.** We do not report robust regression or rank-based alternatives to OLS for the pooled model.

7.3 Conceptual Weaknesses

1. **2D slice fidelity.** The entire signal depends on 2D cross-sections of a high-dimensional surface. Five independent slices reduce sampling variance but do not empirically demonstrate that this approximates the true local basin topology. There is no analysis of slice orientation sensitivity: H_0 in a 2D slice can reflect saddle structure or orientation artifacts rather than true basin multiplicity.

2. **H_0 as capacity proxy.** H_0 is strongly tied to model capacity via the WRN width ladder ($\rho = -1.0$). If capacity correlates with landscape smoothness, which correlates with EWC benefit, then H_0 may simply proxy for scale-dependent regularization dynamics rather than providing independent topological information. The pooled model controls for log(params), but the disentanglement is incomplete.

3. **WRN perfect monotonicity.** $\rho = -1.0$ across all 3 datasets with only 6 data points ($k = 1, 2, 4, 6, 8, 10$) is consistent with a structural relationship but can also occur by chance: the probability of perfect monotonicity in 6 random points is $1/6! \approx 0.0014$ per dataset, yielding $(1/6!)^3 \approx 2.7 \times 10^{-9}$ for all three under independence. The structural interpretation is plausible but not proven.

4. **Mechanistic speculation.** The basin fragmentation hypothesis is entirely correlational. No experiment manipulates landscape fragmentation (e.g., via sharpness-aware minimization) to test whether EWC benefit changes correspondingly.

Falsification targets. Five results that would undermine the hypothesis:

1. Synaptic intelligence benefit shows no H_0 correlation (finding is EWC-specific).

2. Adding 10+ architectures eliminates the CUB-200 topology signal ($p = 0.037$ was a small-sample artifact).

3. SAM changes H_0 without changing EWC benefit (fragmentation–mitigation link is spurious).

4. Cubical persistence disagrees with Ripser on the moderation result (method dependence).

5. A replay-based method shows equal H_0 correlation as EWC (no curvature specificity).

8 Conclusion

Across 19 architectures and 3 datasets (57 configurations), we find that loss landscape topology is a conditional, not universal, predictor of continual learning dynamics. Topology rescues forgetting prediction on CUB-200 (permutation $p = 0.037$, suggestive) where baseline metrics alone are insufficient, but adds no predictive value on CIFAR-100 (where parameter count dominates) or RESISC-45. The WRN-28-k width ladder validates H_0 as a structurally meaningful landscape summary: H_0 monotonically decreases with capacity ($\rho = -1.0$ on all 3 datasets), and Ripser–GUDHI agreement ($\rho = 1.0$) rules out software artifacts, though with only $n = 6$ widths, perfect monotonicity can occur by chance.

In exploratory post-hoc analysis, we additionally observe that H_0 correlates with EWC mitigation benefit on CIFAR-100 ($\rho = 0.76$, 95% CI $[0.46, 0.91]$) and RESISC-45 ($\rho = 0.86$, 95% CI $[0.65, 0.94]$)

but not CUB-200 ($\rho = 0.31$, 95% CI $[-0.22, 0.78]$). A pooled interaction model provides suggestive evidence of dataset-dependent moderation (permutation $p = 0.046$), but this result was not pre-specified, does not survive conservative correction, and should be treated as hypothesis-generating rather than confirmatory. The basin fragmentation hypothesis offers a candidate mechanism linking H_0 to regularization sensitivity, but remains untested.

Critically, these results constitute a *preliminary proof-of-concept* on small-to-medium models (0.3M–44.7M parameters) and small-image datasets. Whether the topological signal survives at production scale remains genuinely unknown and constitutes the primary open research question. Scaling persistent homology to models with hundreds of millions or billions of parameters introduces fundamental computational barriers: Ripser complexity is superpolynomial in the number of sampled points, and it is unclear whether the 2D subsampling strategy retains fidelity in high-dimensional parameter spaces.

Systematic investigation requires HPC resources (e.g., NSF ACCESS supercomputer allocation) and should: (i) test whether the signal persists on production-scale models (100M–7B+ parameters), which may require novel distributed PH algorithms; (ii) extend to long task sequences (10–100+ sequential tasks) where topology evolution may become chaotic; and (iii) evaluate multiple continual learning methods (SI, PackNet, replay, adapter-based) to determine whether the finding generalizes beyond EWC. Additionally, scaling to 50–100+ architectures would provide statistical power sufficient to survive strict multiple-comparison correction, and pre-registering the moderation test would enable confirmatory replication.

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